

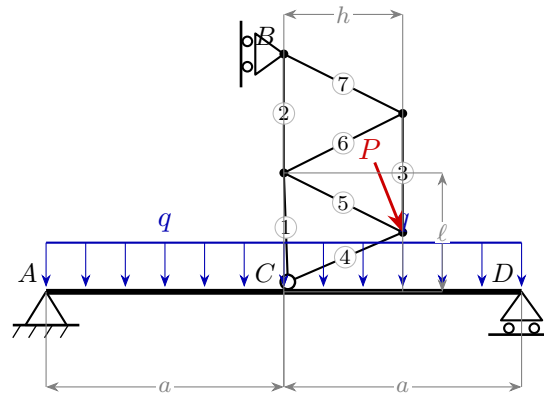
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Exam **Group A**

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

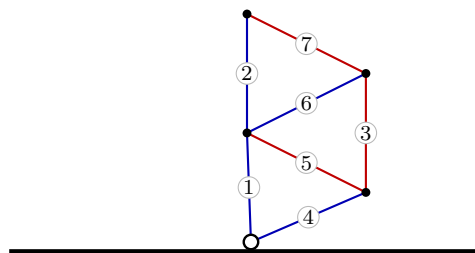
$$A: R_x = 4, R_y = 22 \quad D: R_x = 0, R_y = 22 \quad B: R_x = -12, R_y = 0$$

$$C: H_x = -4, H_y = -20$$

3. Truss member forces

Member	Force S [kN]	Type
1	-18	compression
2	-6	compression
3	12	tension
4	-4.47	compression
5	13.42	tension
6	-13.42	compression
7	13.42	tension

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

